

```
In [97]: import pandas as pd
import numpy as np
df = pd.read_csv(r'C:\Users\KHAN\Desktop\crime.csv')
print(df)
```

	DR_NO	Date Rptd	DATE OCC	TIME OCC	AREA	\
0	190326475	3/1/2020 0:00	3/1/2020 0:00	2130	7	
1	200106753	2/9/2020 0:00	2/8/2020 0:00	1800	1	
2	200320258	11/11/2020 0:00	11/4/2020 0:00	1700	3	
3	200907217	5/10/2023 0:00	3/10/2020 0:00	2037	9	
4	200412582	9/9/2020 0:00	9/9/2020 0:00	630	4	
...	...	...	...	...	...	
1005193	250304214	2/23/2025 0:00	2/21/2025 0:00	1530	3	
1005194	250304203	2/20/2025 0:00	2/13/2025 0:00	2100	3	
1005195	250504051	1/14/2025 0:00	1/14/2025 0:00	1250	5	
1005196	251604136	2/27/2025 0:00	2/27/2025 0:00	1550	16	
1005197	251904210	3/25/2025 0:00	3/25/2025 0:00	1235	19	

	AREA NAME	Rpt Dist No	Part 1-2	Crm Cd	\
0	Wilshire	784	1	510	
1	Central	182	1	330	
2	Southwest	356	1	480	
3	Van Nuys	964	1	343	
4	Hollenbeck	413	1	510	
...	...	...	...	...	
1005193	Southwest	358	1	510	
1005194	Southwest	325	1	522	
1005195	Harbor	509	1	210	
1005196	Foothill	1664	1	510	
1005197	Mission	1944	2	850	

	Crm Cd Desc	...	Status	\
0	VEHICLE - STOLEN	...	AA	
1	BURGLARY FROM VEHICLE	...	IC	
2	BIKE - STOLEN	...	IC	
3	SHOPLIFTING-GRAND THEFT (\$950.01 & OVER)	...	IC	
4	VEHICLE - STOLEN	...	IC	
...	...	...	...	
1005193	VEHICLE - STOLEN	...	IC	
1005194	VEHICLE, STOLEN - OTHER (MOTORIZED SCOOTERS, B...	...	IC	
1005195	ROBBERY	...	IC	
1005196	VEHICLE - STOLEN	...	AA	
1005197	INDECENT EXPOSURE	...	IC	

	Status Desc	Crm Cd 1	Crm Cd 2	Crm Cd 3	Crm Cd 4	\
0	Adult Arrest	510.0	998.0	NaN	NaN	
1	Invest Cont	330.0	998.0	NaN	NaN	

2	Invest	Cont	480.0	NaN	NaN	NaN
3	Invest	Cont	343.0	NaN	NaN	NaN
4	Invest	Cont	510.0	NaN	NaN	NaN
...	...	...	...	...	...	...
1005193	Invest	Cont	510.0	NaN	NaN	NaN
1005194	Invest	Cont	522.0	NaN	NaN	NaN
1005195	Invest	Cont	210.0	NaN	NaN	NaN
1005196	Adult	Arrest	510.0	NaN	NaN	NaN
1005197	Invest	Cont	850.0	NaN	NaN	NaN

				LOCATION	Cross Street	LAT \
0	1900 S	LONGWOOD		AV	NaN	34.0375
1	1000 S	FLOWER		ST	NaN	34.0444
2	1400 W	37TH		ST	NaN	34.0210
3	14000	RIVERSIDE		DR	NaN	34.1576
4			200 E	AVENUE 28	NaN	34.0820
...	...	...	...	...	...	...
1005193	3600	MCCLINTOCK		AV	NaN	34.0212
1005194	2600	ELLENDAL		PL	NaN	34.0307
1005195	24300	WESTERN		AV	NaN	33.8046
1005196	11900	SHELDON		ST	NaN	34.2404
1005197	11100	OMELVENY		AV	NaN	34.2722

	LON
0	-118.3506
1	-118.2628
2	-118.3002
3	-118.4387
4	-118.2130
...	...
1005193	-118.2895
1005194	-118.2923
1005195	-118.3074
1005196	-118.3922
1005197	-118.4417

[1005198 rows x 28 columns]

```
In [12]: df = df[['Date Rptd', 'DATE OCC', 'TIME OCC', 'AREA',
               'Crm Cd Desc', 'Status Desc', 'Vict Age', 'Vict Sex', 'Weapon Desc',
               'Premis Desc']]
df = df.ffill()
```

```
df['Vict Age'] = df['Vict Age'].replace(-4, 4).replace(-3, 3)
               .replace(-2, 2).replace(-1, 1)
df['Vict Age'] = df['Vict Age'].replace(0, pd.NA).ffill()
df['Vict Age'] = df['Vict Age'].fillna(0)
df = df.ffill()
df['Rptd month'] = pd.to_datetime(df['Date Rptd']).dt.month_name()
df['Rptd year'] = pd.to_datetime(df['Date Rptd']).dt.year
df['occ month'] = pd.to_datetime(df['DATE OCC']).dt.month_name()
df['occ year'] = pd.to_datetime(df['DATE OCC']).dt.year
def Vict_Age_Group(m):
    if m < 21: return '0-20'
    elif m < 41: return '20-40'
    elif m < 61: return '40-60'
    elif m < 81: return '60-80'
    else: return '80+'
df['Vict_Age_Group'] = df['Vict Age'].apply(Vict_Age_Group)
df['Time Occur'] = pd.to_datetime(df['TIME OCC']).astype(str)
               .str.zfill(4).format='%H%M').dt.strftime('%I:%M%p')
df = df[['Time Occur', 'Vict_Age_Group', 'occ year', 'occ month',
         'Rptd year', 'Rptd month', 'Premis Desc', 'Weapon Desc', 'Vict Sex',
         'Vict Age', 'Status Desc', 'Crm Cd Desc', 'AREA']]
```

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1005196	Adult Arrest	510.0	NaN	NaN	NaN	
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[1005198 rows x 28 columns]

C:\Users\KHAN\AppData\Local\Temp\ipykernel_8340\2264138855.py:9: FutureWarning: Downcasting object dtype arrays on .fillna, .ffill, .bfill is deprecated and will change in a future version. Call result.infer_objects(copy=False) instead. To opt-in to the future behavior, set `pd.set\_option('future.no\_silent\_downcasting', True)`

df3['Vict Age'] = df3['Vict Age'].fillna(0)

```
In [20]: print(df['Weapon Desc'].unique())
```

```
['STRONG-ARM (HANDS, FIST, FEET OR BODILY FORCE)' 'VERBAL THREAT'
'KNIFE WITH BLADE OVER 6 INCHES IN LENGTH' 'UNKNOWN WEAPON/OTHER WEAPON'
'MACHETE' 'HAND GUN' 'SEMI-AUTOMATIC PISTOL' 'SIMULATED GUN' 'HAMMER'
'RAZOR' 'OTHER FIREARM' 'KITCHEN KNIFE' 'STICK' 'UNKNOWN FIREARM'
'KNIFE WITH BLADE 6INCHES OR LESS' 'PIPE/METAL PIPE'
'AIR PISTOL/REVOLVER/RIFLE/BB GUN' 'OTHER CUTTING INSTRUMENT' 'SCISSORS'
'SWITCH BLADE' 'ROCK/THROWN OBJECT' 'OTHER KNIFE' 'BOTTLE' 'CLUB/BAT'
'BLUNT INSTRUMENT' 'BOARD' 'PHYSICAL PRESENCE' 'RIFLE' 'REVOLVER'
'CLEAVER' 'STUN GUN' 'SCREWDRIVER' 'UNKNOWN TYPE CUTTING INSTRUMENT'
'FOLDING KNIFE' 'VEHICLE' 'MACE/PEPPER SPRAY' 'FIRE'
'CAUSTIC CHEMICAL/POISON' 'SHOTGUN' 'FIXED OBJECT' 'SEMI-AUTOMATIC RIFLE'
'BELT FLAILING INSTRUMENT/CHAIN' 'BRASS KNUCKLES' 'TIRE IRON'
'CONCRETE BLOCK/BRICK' 'AXE' 'SAWED OFF RIFLE/SHOTGUN'
'MARTIAL ARTS WEAPONS' 'DEMAND NOTE' 'DIRK/DAGGER' 'TOY GUN' 'GLASS'
'BOMB THREAT' 'RAZOR BLADE' 'ASSAULT WEAPON/UZI/AK47/ETC'
'HECKLER & KOCH 93 SEMIAUTOMATIC ASSAULT RIFLE' 'EXPLOXIVE DEVICE'
'SCALDING LIQUID' 'MAC-11 SEMIAUTOMATIC ASSAULT WEAPON' 'SWORD'
'LIQUOR/DRUGS' 'BOW AND ARROW' 'DOG/ANIMAL (SIC ANIMAL ON)' 'ICE PICK'
'ROPE/LIGATURE' 'ANTIQUE FIREARM' 'BOWIE KNIFE' 'SYRINGE'
'AUTOMATIC WEAPON/SUB-MACHINE GUN' 'M1-1 SEMIAUTOMATIC ASSAULT RIFLE'
'STARTER PISTOL/REVOLVER' 'STRAIGHT RAZOR'
'UZI SEMIAUTOMATIC ASSAULT RIFLE' 'MAC-10 SEMIAUTOMATIC ASSAULT WEAPON'
'BLACKJACK' 'RELIC FIREARM'
'HECKLER & KOCH 91 SEMIAUTOMATIC ASSAULT RIFLE'
'UNK TYPE SEMIAUTOMATIC ASSAULT RIFLE' 'M-14 SEMIAUTOMATIC ASSAULT RIFLE']
```

```
In [8]: df['Main_weapon'] = np.where(df['Weapon Desc'].str
    .contains('STRONG-ARM|PHYSICAL PRESENCE|VERBAL THREAT|ROPE|DOG|DRUGS',
    na=False), 'Physical Force', df['Weapon Desc'])

df['Main_weapon'] = np.where(df['Main_weapon'].str
    .contains('CLUB|STICK|PIPE|HAMMER|ROCK|BRICK|IRON|BRASS|SCREWDRIVER|BLUNT|SCISSORS',
    na=False), 'Blunt Instruments', df['Main_weapon'])

df['Main_weapon'] = np.where(df['Main_weapon'].str
    .contains('KNIFE|BLADE|CLEAVER|RAZOR|DIRK|SWORD|AXE|BOW|UNKNOWN TYPE CUTTING
    |ICE PICK|MACHETE', na=False), 'Sharp instruments', df['Main_weapon'])

df['Main_weapon'] = np.where(df['Main_weapon'].str
    .contains('GUN|RIFLE|PISTOL|REVOLVER|MAC-10|FIREARM|ASSAULT', na=False),
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    'Fire arms', df['Main_weapon'])
```

```
df['Main_weapon'] = np.where(df['Main_weapon'].str
    .contains('BOMB|EXPLOSIVE|FIRE|LIQUID|POISON|SPRAY|SYRINGE|EXPLOXIVE', na=False),
    'Explosive & Chemical threat', df['Main_weapon'])

df['Main_weapon'] = np.where(df['Main_weapon'].str
    .contains('VEHICLE|MARTIAL ARTS|DEMAND NOTE|GLASS|BOARD|OTHER|CHAIN|OBJECT|BELT|
    BLACKJACK|BOTTLE', na=False), 'Other', df['Main_weapon'])
```

```
In [12]: print(df['Main_weapon'].unique())
```

```
['Physical Force' 'Sharp instruments' 'Other' 'Fire arms'
'Blunt Instruments' 'Explosive & Chemical threat']
```

```
In [16]: print(df['Premis Desc'].unique())
```

['MULTI-UNIT DWELLING (APARTMENT, DUPLEX, ETC)' 'ALLEY' 'STREET'
'GAS STATION' 'CAR WASH' 'SINGLE FAMILY DWELLING' 'CONDOMINIUM/TOWNHOUSE'
'PARKING LOT' 'OTHER BUSINESS' 'RESTAURANT/FAST FOOD' 'SIDEWALK'
'NURSING/CONVALESCENT/RETIREMENT HOME' 'MARKET'
'GOVERNMENT FACILITY (FEDERAL,STATE, COUNTY & CITY)' 'DRIVEWAY'
'MINI-MART' 'YARD (RESIDENTIAL/BUSINESS)' 'VEHICLE, PASSENGER/TRUCK'
'GARAGE/CARPORT' 'PARKING UNDERGROUND/BUILDING' 'NAIL SALON'
'PORCH, RESIDENTIAL' 'DRUG STORE' 'MTA BUS'
'MTA - RED LINE - VERMONT/BEVERLY' 'OTHER PREMISE' 'DAY CARE/CHILDREN*'
'BUS STOP' 'POLICE FACILITY' 'MISSIONS/SHELTERS'
'CHURCH/CHAPEL (CHANGED 03-03 FROM CHURCH/TEMPLE)'
'PHARMACY INSIDE STORE OR SUPERMARKET*'
'DISCOUNT STORE (99 CENT,DOLLAR,ETC.)' 'PUBLIC STORAGE'
'MTA - RED LINE - 7TH AND METRO CENTER' 'OFFICE BUILDING/OFFICE'
'OTHER STORE' 'DELIVERY SERVICE (FED EX, UPS, COURIERS,COURIER SERVICE)*'
'HOTEL' 'MORTUARY' 'OTHER RESIDENCE' 'OTHER/OUTSIDE'
'MTA - EXPO LINE - EXPO/LA BREA' 'JUNIOR HIGH SCHOOL'
'NIGHT CLUB (OPEN EVENINGS ONLY)'
'COFFEE SHOP (STARBUCKS, COFFEE BEAN, PEET'S, ETC.)" 'CYBERSPACE'
'HIGH SCHOOL' 'CONSTRUCTION SITE' 'STUDIO (FILM/PHOTOGRAPHIC/MUSIC)'
'BEAUTY/BARBER SHOP' 'MOTEL' 'VALET' 'PARK/PLAYGROUND'
'TRANSITIONAL HOUSING/HALFWAY HOUSE' 'LAUNDROMAT' 'AUTO SALES LOT'
'ELEMENTARY SCHOOL' 'LA UNION STATION (NOT LINE SPECIFIC)'
'MTA - EXPO LINE - EXPO/VERMONT' 'BUS DEPOT/TERMINAL, OTHER THAN MTA'
'SPECIALTY SCHOOL/OTHER' 'CONVENTION CENTER' 'DETENTION/JAIL FACILITY'
'DIY CENTER (LOWE'S,HOME DEPOT,OSH,CONTRACTORS WAREHOUSE)'
'NURSERY/FLOWER SHOP' 'GROUP HOME' 'RECYCLING CENTER'
'7TH AND METRO CENTER (NOT LINE SPECIFIC)'
'FRAT HOUSE/SORORITY/DORMITORY' 'TAXI' 'CLOTHING STORE' 'HOSPITAL'
'DEPARTMENT STORE' 'PROJECT/TENEMENT/PUBLIC HOUSING'
'TRANSIENT ENCAMPMENT' 'MTA - RED LINE - WESTLAKE/MACARTHUR PARK'
'THE GROVE' "SINGLE RESIDENCE OCCUPANCY (SRO'S) LOCATIONS"
'COLLEGE/JUNIOR COLLEGE/UNIVERSITY' 'VETERINARIAN/ANIMAL HOSPITAL'
'CHECK CASHING*' 'UNDERPASS/BRIDGE*' 'HEALTH SPA/GYM' 'MAIL BOX'
'MEDICAL/DENTAL OFFICES' 'LIQUOR STORE' 'AUTO REPAIR SHOP' 'TUNNEL'
'AUTO SUPPLY STORE*' 'MTA - EXPO LINE - EXPO/WESTERN'
'SHOPPING MALL (COMMON AREA)' 'BEAUTY SUPPLY STORE' 'DRIVE THRU*'
'THE BEVERLY CONNECTION' 'PUBLIC RESTROOM(INDOORS-INSIDE)'
'JEWELRY STORE' 'MTA - GOLD LINE - UNION STATION' 'POST OFFICE'
'APARTMENT/CONDO COMMON LAUNDRY ROOM' 'SYNAGOGUE/TEMPLE' 'MASSAGE PARLOR'
'MUNICIPAL BUS LINE INCLUDES LADOT/DASH'
'ABANDONED BUILDING ABANDONED HOUSE' 'PAWN SHOP'

'HARDWARE/BUILDING SUPPLY' 'STAPLES CENTER *'
'MTA - BLUE LINE - WASHINGTON' 'LIBRARY' 'FURNITURE STORE' 'STAIRWELL*'
'STORAGE SHED' 'HIGH-RISE BUILDING' 'THE BEVERLY CENTER'
'TV/RADIO/APPLIANCE' 'TOBACCO SHOP' 'MTA - RED LINE - UNION STATION'
'PATIO*' 'CELL PHONE STORE' 'MUSEUM'
'MTA - PURPLE LINE - CIVIC CENTER/GRAND PARK' 'CLEANER/LAUNDROMAT'
'MTA - RED LINE - CIVIC CENTER/GRAND PARK'
'AUTOMATED TELLER MACHINE (ATM)' 'SHORT-TERM VACATION RENTAL' 'PET STORE'
'MTA - GOLD LINE - HIGHLAND PARK' 'ELEVATOR'
'MOBILE HOME/TRAILERS/CONSTRUCTION TRAILERS/RV'S/MOTORHOME"
'DRIVE THRU BANKING (WINDOW)*' 'FREEWAY' 'BAR/COCKTAIL/NIGHTCLUB'
'MEMBERSHIP STORE (COSTCO,SAMS CLUB)*' 'BANK' 'DAY CARE/ADULTS*'
'MTA - RED LINE - PERSHING SQUARE' 'SLIPS/DOCK/MARINA/BOAT' 'BALCONY*'
'THEATRE/MOVIE'
'VEHICLE STORAGE LOT (CARS, TRUCKS, RV'S, BOATS, TRAILERS, ETC.)"
'CEMETARY*' 'MTA - EXPO LINE - FARMDALE' 'PRIVATE SCHOOL/PRESCHOOL'
'CREDIT UNION' 'WAREHOUSE' 'OIL REFINERY'
'SKATEBOARD FACILITY/SKATEBOARD PARK*' 'MTA - RED LINE - VERMONT/SUNSET'
'BEACH' 'TRAIN TRACKS' 'GUN/SPORTING GOODS' 'COLISEUM'
'MEDICAL MARIJUANA FACILITIES/BUSINESSES' 'WEBSITE'
'FOSTER HOME BOYS OR GIRLS*' 'MTA - RED LINE - VERMONT/SANTA MONICA'
'MTA - EXPO LINE - EXPO/SEPULVEDA'
'ELECTRONICS STORE (IE:RADIO SHACK, ETC.)' 'TOW YARD*'
'VISION CARE FACILITY*' 'TRASH CAN/TRASH DUMPSTER'
'TRANSPORTATION FACILITY (AIRPORT)' 'MTA - RED LINE - HOLLYWOOD/HIGHLAND'
'AIRCRAFT' "SEX ORIENTED/BOOK STORE/STRIP CLUB/GENTLEMAN'S CLUB"
'BAR/SPORTS BAR (OPEN DAY & NIGHT)'
'AUTO DEALERSHIP (CHEVY, FORD, BMW, MERCEDES, ETC.)'
'PEDESTRIAN OVERCROSSING'
'MTA - SILVER LINE - HARBOR GATEWAY TRANSIT CTR' 'BASKETBALL COURTS'
'MTA - EXPO LINE - 7TH AND METRO CENTER'
'ENTERTAINMENT/COMEDY CLUB (OTHER)' 'MTA - EXPO LINE - EXPO/BUNDY'
'GOLF COURSE*' 'FIRE STATION' 'TELECOMMUNICATION FACILITY/LOCATION'
'FACTORY' 'MTA - RED LINE - HOLLYWOOD/VINE' 'SPORTS VENUE, OTHER'
'PUBLIC RESTROOM/OUTSIDE*' 'MTA - PURPLE LINE - WESTLAKE/MACARTHUR PARK'
'HOSPICE' 'MTA - BLUE LINE - SAN PEDRO' 'BUS-CHARTER/PRIVATE'
'RIVER BED*' 'MTA - ORANGE LINE - RESEDA' 'MTA - SILVER LINE - ROSECRANS'
'MANUFACTURING COMPANY' 'TATTOO PARLOR*'
'OPTICAL OFFICE INSIDE STORE OR SUPERMARKET*'
'MTA - EXPO LINE - EXPO PARK/USC' 'MTA - SILVER LINE - HARBOR FWY'
'MTA - ORANGE LINE - TAMPA' 'POOL-PUBLIC/OUTDOOR OR INDOOR*'
'METROLINK TRAIN' 'MTA - EXPO LINE - EXPO/CRENSHAW'

'MTA - EXPO LINE - LA CIENEGA/JEFFERSON' 'EQUIPMENT RENTAL' 'VACANT LOT'
'MTA - BLUE LINE - GRAND/LATTC' 'MTA - BLUE LINE - PICO'
'BANKING INSIDE MARKET-STORE *' 'AMUSEMENT PARK*'
'MTA - EXPO LINE - JEFFERSON/USC' 'BUS STOP/LAYOVER (ALSO QUERY 124)'
'SPORTS ARENA' 'MTA - BLUE LINE - 7TH AND METRO CENTER'
'MTA - RED LINE - HOLLYWOOD/WESTERN' 'SWAP MEET' 'BUS, SCHOOL, CHURCH'
'MTA - GOLD LINE - SOUTHWEST MUSEUM' 'GREYHOUND OR INTERSTATE BUS'
'TRADE SCHOOL (MEDICAL-TECHNICAL-BUSINESS)*'
'MTA - PURPLE LINE - UNION STATION' 'MTA - GOLD LINE - SOTO'
'MTA - BLUE LINE - 103RD/WATTS TOWERS' 'BOWLING ALLEY*'
'MTA - ORANGE LINE - WOODMAN' 'MTA - EXPO LINE - PICO'
'MTA - ORANGE LINE - VAN NUYS' 'OTHER PLACE OF WORSHIP'
'MTA PROPERTY OR PARKING LOT' 'MTA - PURPLE LINE - 7TH AND METRO CENTER'
'OTHER RR TRAIN (UNION PAC, SANTE FE ETC'
'BANK DROP BOX/MONEY DROP-OUTSIDE OF BANK*'
'COMPUTER SERVICES/REPAIRS/SALES' 'MTA - GOLD LINE - LINCOLN/CYPRESS'
'ABATEMENT LOCATION' 'MTA - GOLD LINE - MARIACHI PLAZA'
'MTA - EXPO LINE - WESTWOOD/RANCHO PARK' 'BOOK STORE' 'SAVINGS & LOAN'
'MTA - SILVER LINE - SLAUSON' 'MTA - ORANGE LINE - BALBOA'
'ABORTION CLINIC/ABORTION FACILITY*'
'TRAIN DEPOT/TERMINAL, OTHER THAN MTA'
'MTA - PURPLE LINE - PERSHING SQUARE' 'MTA - ORANGE LINE - WOODLEY'
'MTA - GOLD LINE - HERITAGE SQ' 'RECORD-CD MUSIC/COMPUTER GAME STORE'
'MTA - BLUE LINE - VERNON' 'MTA - SILVER LINE - UNION STATION'
'SEWAGE FACILITY/PIPE' 'DODGER STADIUM' 'MTA - SILVER LINE - 37TH ST/USC'
'MTA - GOLD LINE - CHINATOWN' 'MTA - GOLD LINE - INDIANA' 'TOOL SHED*'
'METHADONE CLINIC' 'PAY PHONE' 'AMTRAK TRAIN'
'MTA - SILVER LINE - DOWNTOWN STREET STOPS' 'VIDEO RENTAL STORE'
'TRUCK, COMMERICAL' 'MTA - SILVER LINE - PACIFIC COAST HWY'
'SURPLUS SURVIVAL STORE' 'HORSE RACING/SANTA ANITA PARK*'
'CATERING/ICE CREAM TRUCK' 'FINANCE COMPANY' 'DAM/RESERVOIR'
'CULTURAL SIGNIFICANCE/MONUMENT' 'MTA - ORANGE LINE - SEPULVEDA'
'MTA - GOLD LINE - LITTLE TOKYO/ARTS DISTRICT'
'MTA - GOLD LINE - PICO/ALISO' 'MASS GATHERING LOCATION' 'WATER FACILITY'
'OTHER INTERSTATE, CHARTER BUS' 'MTA - EXPO LINE - LATTC/ORTHO INSTITUTE'
'MTA - ORANGE LINE - VALLEY COLLEGE' 'ESCALATOR*'
'MTA - ORANGE LINE - SHERMAN WAY' 'MTA - ORANGE LINE - NORTH HOLLYWOOD'
'MTA - RED LINE - NORTH HOLLYWOOD'
'MTA - PURPLE LINE - WILSHIRE/NORMANDIE' 'ENERGY PLANT/FACILITY'
'MTA - ORANGE LINE - CHATSWORTH' 'MTA - PURPLE LINE - WILSHIRE/WESTERN'
'MTA - RED LINE - UNIVERSAL CITY/STUDIO CITY'
'MTA - RED LINE - WILSHIRE/VERMONT' 'MTA - GREEN LINE - AVALON'

'MTA - PURPLE LINE - WILSHIRE/VERMONT' 'MTA - GREEN LINE - HARBOR FWY'
'MTA - EXPO LINE - PALMS' 'SKATING RINK*' 'HANDBALL COURTS'
'MTA - ORANGE LINE - CANOGA' 'MTA - ORANGE LINE - NORDHOFF'
'TERMINAL, OTHER THAN MTA' 'MUSCLE BEACH'
'MTA - GREEN LINE - AVIATION/LAX'
'ARCADE,GAME ROOM/VIDEO GAMES (EXAMPLE CHUCKIE CHEESE)*'
'GARMENT MANUFACTURER' 'MTA - ORANGE LINE - LAUREL CANYON'
'MTA - ORANGE LINE - PIERCE COLLEGE' 'MTA - ORANGE LINE - ROSCOE'
'MOSQUE*' 'MTA - ORANGE LINE - DE SOTO'
'TRAM/STREETCAR(BOXLIKE WAG ON RAILS)*'
'RETIRED (DUPLICATE) DO NOT USE THIS CODE'
'CHEMICAL STORAGE/MANUFACTURING PLANT'
'MTA - SILVER LINE - LAC/USC MEDICAL CENTER' 'DEPT OF DEFENSE FACILITY'
'TRAIN, OTHER THAN MTA (ALSO QUERY 809/810/811)' 'HOCKEY RINK/ICE HOCKEY'
'MTA - SILVER LINE - MANCHESTER'
'HARBOR FRWY STATION (NOT LINE SPECIFIC)']

```
In [23]: df['Main_Premis'] = np.where(df['Premis_Desc'].str
        .contains('HOUSE|APARTMENT|DWELLING|RESIDENCE|BUILDING|HOME|
        'YARD|CARPORT|GARAGE|SHED|DORMITORY|SRO|HOUSING|INDOOR|PATIO*|RESIDENTIAL|BALCONY*
        '|PORCH|TRANSIENT', na=False), 'Residencial Locations', df['Premis_Desc'])

df['Main_Premis'] = np.where(df['Main_Premis'].str
        .contains('RESTAURANT|BAR|SHOP|NIGHT CLUB|FOOD|MARKET|STORE|MART|PHARMACY|SALON|MALL
        '|AUTO|CENTER|STUDIO|BUSINESS|BANK|MAIL BOX|CHECK CASHING|COMPUTER|SAVING|CLINIC
        '|OUTSIDE|SWAP|PAY PHONE|WASH|GROVE|AIRCRAFT|UNION', na=False),
        'Commercial Location', df['Main_Premis'])

df['Main_Premis'] = np.where(df['Main_Premis'].str
        .contains('FACILITY|HOSPITAL|NURSING|HOSPICE|STATION|DETENTION|DENTAL|SCHOOL|COLLEGE
        '|UNIVERSITY|MISSIONS|SHELTERS|DAY CARE|MUSEUM|POST|LIBRARY|MISSION|BANKING|
        'RESTROOM|VATEL|COLISEUM|BEVERLY|CLEANER|LAUN', na=False), 'Public Facilities',
        df['Main_Premis'])

df['Main_Premis'] = np.where(df['Main_Premis'].str
        .contains('STREET|SIDEWALK|PARKING|VEHICLE|TOW|STOP|DEPOT|LINE|BUS|TRAIN|
        'MTA|AIRPORT|TAXI|TRUCK|TUNNEL|FREEWAY|UNDERPASS|CROSSING|MOTEL|DELIVERY|
        'RIVER', na=False), 'Transportation', df['Main_Premis'])

df['Main_Premis'] = np.where(df['Main_Premis'].str
        .contains('PARK|BEACH|VACANT|PLAY|SPORT|STADIUM|CLUB|THEATRE|ARCADE|ALLEY|RINK
        '|GOLF|BASKETBALL|HANDBALL|GATHERING|PARLOR|HOTEL|BOAT|HEALTH|TV', na=False),
```

```

        'Entertainment Site', df['Main_Premis'])

df['Main_Premis'] = np.where(df['Main_Premis'].str.
    contains('COMPANY|MANUFACTURER|FACTORY|PLANT|REFINERY|RECYCLE|ENERGY|PUBLIC STORAGE|
    'WAREHOUSE|RENTAL|SITE|TELECOM|ABATEMENT|DAM|CYBERSPACE', na=False),
    'Industerial Site', df['Main_Premis'])

df['Main_Premis'] = np.where(df['Main_Premis'].str
    .contains('CHURCH|TEMPLE|MOSQUE|WORSHIP|CEMETERY|MISSIONS|TRADITIONAL|CULTURAL|
    'COMMUNITY|MORTUARY|CEMETARY', na=False), 'Regilious Site', df['Main_Premis'])

df['Main_Premis'] = np.where(df['Main_Premis'].str
    .contains('STAIRWELL|TRASH CAN/TRASH DUMPSTER|ELEVATOR|RETIRED|OTHER|VALET|
    'ESCALATOR*', na=False), 'Other', df['Main_Premis'])

In [15]: print(df['Main_Premis'].unique())

['Residencial Locations' 'Entertainment Site' 'Transportation'
'Public Facilities' 'Commercial Location' 'Other' 'Regilious Site'
'Industerial Site']

In [2]: print(df['Crm Cd Desc'].unique())

```

```

['INTIMATE PARTNER - SIMPLE ASSAULT' 'BURGLARY FROM VEHICLE' 'ROBBERY'
'BURGLARY' 'THEFT-GRAND ($950.01 & OVER)EXCPT,GUNS,FOWL,LIVESTK,PROD'
'ASSAULT WITH DEADLY WEAPON, AGGRAVATED ASSAULT' 'THEFT OF IDENTITY'
'BATTERY - SIMPLE ASSAULT' 'SHOPLIFTING - PETTY THEFT ($950 & UNDER)'
'BUNCO, GRAND THEFT' 'SHOPLIFTING-GRAND THEFT ($950.01 & OVER)'
'VIOLATION OF COURT ORDER' 'VEHICLE - STOLEN'
'VIOLATION OF RESTRAINING ORDER' 'THEFT PLAIN - PETTY ($950 & UNDER)'
'VANDALISM - FELONY ($400 & OVER, ALL CHURCH VANDALISMS)'
'RAPE, FORCIBLE' 'THEFT FROM MOTOR VEHICLE - GRAND ($950.01 AND OVER)'
'BIKE - STOLEN' 'TRESPASSING' 'VANDALISM - MISDEAMEANOR ($399 OR UNDER)'
'VEHICLE - ATTEMPT STOLEN' 'RESISTING ARREST'
'EMBEZZLEMENT, GRAND THEFT ($950.01 & OVER)'
'BURGLARY FROM VEHICLE, ATTEMPTED'
'LETTERS, LEWD - TELEPHONE CALLS, LEWD'
'CRIMINAL THREATS - NO WEAPON DISPLAYED'
'SEX OFFENDER REGISTRANT OUT OF COMPLIANCE'
'UNAUTHORIZED COMPUTER ACCESS'
'THEFT FROM MOTOR VEHICLE - PETTY ($950 & UNDER)'
'CRM AGNST CHLD (13 OR UNDER) (14-15 & SUSP 10 YRS OLDER)'
'BRANDISH WEAPON' 'BURGLARY, ATTEMPTED' 'DISCHARGE FIREARMS/SHOTS FIRED'
'BATTERY POLICE (SIMPLE)'
'VEHICLE, STOLEN - OTHER (MOTORIZED SCOOTERS, BIKES, ETC)'
'ORAL COPULATION' 'INDECENT EXPOSURE' 'THEFT FROM PERSON - ATTEMPT'
'CHILD ABUSE (PHYSICAL) - SIMPLE ASSAULT' 'OTHER ASSAULT'
'DISTURBING THE PEACE' 'INTIMATE PARTNER - AGGRAVATED ASSAULT'
'OTHER MISCELLANEOUS CRIME' 'BOMB SCARE' 'FAILURE TO YIELD'
'CONTEMPT OF COURT' 'ATTEMPTED ROBBERY'
'ASSAULT WITH DEADLY WEAPON ON POLICE OFFICER'
'DOCUMENT FORGERY / STOLEN FELONY' 'BUNCO, PETTY THEFT'
'SEXUAL PENETRATION W/FOREIGN OBJECT' 'SHOTS FIRED AT INHABITED DWELLING'
'CHILD STEALING' 'DEFRAUDING INNKEEPER/THEFT OF SERVICES, $950 & UNDER'
'KIDNAPPING - GRAND ATTEMPT'
'SHOTS FIRED AT MOVING VEHICLE, TRAIN OR AIRCRAFT' 'THEFT, PERSON'
'CHILD ABUSE (PHYSICAL) - AGGRAVATED ASSAULT' 'ARSON' 'EXTORTION'
'CHILD NEGLECT (SEE 300 W.I.C.)'
'TILL TAP - GRAND THEFT ($950.01 & OVER)'
'SEX,UNLAWFUL(INC MUTUAL CONSENT, PENETRATION W/ FRGN OBJ'
'BATTERY WITH SEXUAL CONTACT' 'HUMAN TRAFFICKING - COMMERCIAL SEX ACTS'
'CHILD ANNOYING (17YRS & UNDER)' 'DOCUMENT WORTHLESS ($200.01 & OVER)'
'RAPE, ATTEMPTED' 'FALSE IMPRISONMENT'
'THROWING OBJECT AT MOVING VEHICLE' 'LEWD CONDUCT' 'PEEPING TOM'
'KIDNAPPING' 'CRIMINAL HOMICIDE' 'STALKING' 'THEFT PLAIN - ATTEMPT'

```

'SODOMY/SEXUAL CONTACT B/W PENIS OF ONE PERS TO ANUS OTH'
 'VIOLATION OF TEMPORARY RESTRAINING ORDER' 'CHILD PORNOGRAPHY'
 'WEAPONS POSSESSION/BOMBING' 'DRIVING WITHOUT OWNER CONSENT (DWOC)'
 'THEFT FROM MOTOR VEHICLE - ATTEMPT' 'PICKPOCKET' 'SHOPLIFTING - ATTEMPT'
 'COUNTERFEIT' 'BUNCO, ATTEMPT'
 'DEFRAUDING INNKEEPER/THEFT OF SERVICES, OVER \$950.01'
 'CRUELTY TO ANIMALS' 'FALSE POLICE REPORT' 'PROWLER'
 'DISHONEST EMPLOYEE - GRAND THEFT' 'THREATENING PHONE CALLS/LETTERS'
 'PURSE SNATCHING' 'PANDERING' 'EMBEZZLEMENT, PETTY THEFT (\$950 & UNDER)'
 'PIMPING' 'DOCUMENT WORTHLESS (\$200 & UNDER)' 'ILLEGAL DUMPING'
 'LEWD/LASCIVIOUS ACTS WITH CHILD' 'BATTERY ON A FIREFIGHTER'
 'PETTY THEFT - AUTO REPAIR' 'MANSLAUGHTER, NEGLIGENT' 'RECKLESS DRIVING'
 'TILL TAP - PETTY (\$950 & UNDER)' 'PURSE SNATCHING - ATTEMPT'
 'LYNCHING - ATTEMPTED' 'CREDIT CARDS, FRAUD USE (\$950.01 & OVER)'
 'CREDIT CARDS, FRAUD USE (\$950 & UNDER)'
 'THEFT, COIN MACHINE - PETTY (\$950 & UNDER)'
 'HUMAN TRAFFICKING - INVOLUNTARY SERVITUDE' 'BIKE - ATTEMPTED STOLEN'
 'CONTRIBUTING' 'BRIBERY' 'BOAT - STOLEN' 'CONSPIRACY'
 'GRAND THEFT / INSURANCE FRAUD' 'DRUGS, TO A MINOR' 'CHILD ABANDONMENT'
 'THEFT, COIN MACHINE - GRAND (\$950.01 & OVER)' 'DISRUPT SCHOOL'
 'THEFT, COIN MACHINE - ATTEMPT' 'DISHONEST EMPLOYEE - PETTY THEFT'
 'LYNCHING' 'FIREARMS RESTRAINING ORDER (FIREARMS RO)'
 'REPLICA FIREARMS(SALE,DISPLAY,MANUFACTURE OR DISTRIBUTE)'
 'GRAND THEFT / AUTO REPAIR' 'DRUNK ROLL' 'PICKPOCKET, ATTEMPT'
 'TELEPHONE PROPERTY - DAMAGE'
 'BEASTIALITY, CRIME AGAINST NATURE SEXUAL ASSLT WITH ANIM' 'BIGAMY'
 'FAILURE TO DISPERSE'
 'FIREARMS EMERGENCY PROTECTIVE ORDER (FIREARMS EPO)'
 'INCEST (SEXUAL ACTS BETWEEN BLOOD RELATIVES)'
 'BLOCKING DOOR INDUCTION CENTER' 'INCITING A RIOT'
 'DISHONEST EMPLOYEE ATTEMPTED THEFT' 'TRAIN WRECKING'
 'DRUNK ROLL - ATTEMPT']

```
In [9]: df['Main_Crime'] = np.where(df['Crm Cd Desc'].str
      .contains('ASSAULT|INTIMATE|FIREFIGHTER|RESISTING ARREST|BATTERY', na=False)
      , 'Crimes Against Person', df['Crm Cd Desc'])

df['Main_Crime'] = np.where(df['Crm Cd Desc'].str
      .contains('RAPE|SODOMY|SEXUAL|FORIEGN|PEEPING|INDECENT|LEWD|PORNOGRAPHY|LASCIVIOUS|
      'INCEST|SEX|SEX ACTS|PIMPING|PANDERING|BEASTIALITY|ORAL', na=False),
      'Sexual & Exploitation Crimes', df['Main_Crime'])
```

```
df['Main_Crime'] = np.where(df['Crm Cd Desc'].str
      .contains('BURGLARY|ROBBERY|SHOPLIFTING|THEFT|SNATCHING|PICKPOCKET|STOLEN|BOAT|
      'BIKE|DAMAGE|VANDALISM|THROWING', na=False), 'Property Crimes', df['Main_Crime'])

df['Main_Crime'] = np.where(df['Crm Cd Desc'].str
      .contains('IDENTITY|BUNCO|EMBEZZLEMENT|FRAUD|FORGERY|WORTHLESS|GRAND THEFT|
      DEFRAUDING|THEFT OF SERVICES|DISHONEST|CREDIT CARDS|PETTY|COUNTERFEIT', na=False)
      , 'Fraud & Financial crimes', df['Main_Crime'])

df['Main_Crime'] = np.where(df['Crm Cd Desc'].str
      .contains('DISTURBING|FAILURE|ROIT|CONSPIRACY|COURT|ORDER|STALKING|CRIMINAL THREATS
      '|CALL|LETTER|TRESPASSING|PROWLER|CONTRIBUTING|DUMPING|PROPERTY DAMAGE|OTHER|
      POLICE REPORT|BLOCKING DOOR|DISRUPT SCHOOL', na=False),
      'Public Order & Miscellaneous', df['Main_Crime'])

df['Main_Crime'] = np.where(df['Crm Cd Desc'].str
      .contains('HOMICIDE|MANSLAUGHTER|KIDNAPPING|CHILD STEALING|IMPRISONMENT|BRIBERY|
      'EXTORTION|ARSON|BOMB|WRECKING|BOMBING|WEAPONS|BRANDISH WEAPON|FIREARMS|SHOTS FIRED|
      'LYNCHING|HOMICIDE|TRAFFICKING', na=False), 'Serious Crimes', df['Main_Crime'])

df['Main_Crime'] = np.where(df['Crm Cd Desc'].str
      .contains('SEX OFFENDER|UNAUTHORIZED COMPUTER|DRUGS|ABUSE|ABANDONMENT|ANNOYING
      '|CRUELTY TO ANIMALS|BIGAMY|DRUNK ROLL|BEASTIALITY|AGAINST NATURE|NEGLECT|
      'DRIVING|RIOT|CRM', na=False), 'Others', df['Main_Crime'])
```

```
In [10]: print(df['Main_Crime'].unique())

['Crimes Against Person' 'Property Crimes' 'Fraud & Financial crimes'
'Public Order & Miscellaneous' 'Sexual & Exploitation Crimes' 'Others'
'Serious Crimes']
```

```
In [12]: df2 = df[['Vict_Age_Group', 'occ year', 'occ month', 'Rptd year', 'Rptd month',
      'Vict Sex', 'Status Desc', 'AREA', 'Main_weapon', 'Main_Premis', 'Main_Crime']]
```

```
In [16]: df2['Vict Sex'] = df['Vict Sex'].str.strip().str.title()
df2['Status Desc'] = df['Status Desc'].str.strip().str.title()
df2['Main_weapon'] = df['Main_weapon'].str.strip().str.title()
df2['Main_Crime'] = df['Main_Crime'].str.strip().str.title()
```

```

C:\Users\KHAN\AppData\Local\Temp\ipykernel_7080\1883438895.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
-a-view-versus-a-copy
df2['Vict Sex'] = df['Vict Sex'].str.strip().str.title()
C:\Users\KHAN\AppData\Local\Temp\ipykernel_7080\1883438895.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
-a-view-versus-a-copy
df2['Status Desc'] = df['Status Desc'].str.strip().str.title()
C:\Users\KHAN\AppData\Local\Temp\ipykernel_7080\1883438895.py:3: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
-a-view-versus-a-copy
df2['Main_weapon'] = df['Main_weapon'].str.strip().str.title()
C:\Users\KHAN\AppData\Local\Temp\ipykernel_7080\1883438895.py:4: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning
-a-view-versus-a-copy
df2['Main_Crime'] = df['Main_Crime'].str.strip().str.title()

```

```
In [17]: print(df2)
```

	Vict_Age_Group	occ year	occ month	Rptd year	Rptd month	Vict Sex \
0	20-40	2020	May	2020	May	F
1	20-40	2020	March	2020	March	M
2	20-40	2020	December	2020	December	M
3	0-20	2020	April	2020	April	X
4	0-20	2020	June	2020	June	X
...	...	...	...	...	...	...
1005149	40-60	2025	February	2025	February	M
1005150	40-60	2025	February	2025	February	M
1005151	0-20	2025	January	2025	January	F
1005152	0-20	2025	February	2025	February	F
1005153	20-40	2025	March	2025	March	F

	Status Desc	AREA	Main_weapon	Main_Premis \
0	Adult Arrest	1	Physical Force	Residencial Locations
1	Invest Cont	1	Physical Force	Entertainment Site
2	Invest Cont	12	Physical Force	Transportation
3	Invest Cont	3	Physical Force	Public Facilities
4	Invest Cont	2	Physical Force	Commercial Location
...	...	...	...	...
1005149	Invest Cont	3	Sharp Instruments	Transportation
1005150	Invest Cont	3	Sharp Instruments	Transportation
1005151	Invest Cont	5	Physical Force	Public Facilities
1005152	Adult Arrest	16	Physical Force	Transportation
1005153	Invest Cont	19	Physical Force	Public Facilities

	Main_Crime
0	Crimes Against Person
1	Property Crimes
2	Property Crimes
3	Property Crimes
4	Property Crimes
...	...
1005149	Property Crimes
1005150	Public Order & Miscellaneous
1005151	Property Crimes
1005152	Property Crimes
1005153	Sexual & Exploitation Crimes

```
[1005154 rows x 11 columns]
```

```
In [18]: import os
desktop = os.path.join(os.path.expanduser('~'), 'Desktop')
df2.to_csv(r'C:\Users\KHAN\Desktop\crimereport.csv')
```